

Name: Priyanshu Digari
Roll no. : 89

Assignment -07
Image Processing and Pattern
Recognition
TCS-361

Q1.) Define nonparametric techniques in pattern recognition. How are they different from parametric methods?

Ans.) Nonparametric techniques in pattern recognition are methods that do not assume a specific form for the underlying data distribution.

Examples: KNN, Parzen window.

Difference from Parametric Methods:

- Parametric: Assume fixed distribution (e.g., Gaussian), limited

parameters

- Nonparametric: No assumption about distribution
- Parametric: Faster, less data required
- Nonparametric: More flexible, needs more data

Q2.) Explain Density Estimation and discuss its importance in classification problems.

Ans.) Density estimation is the process of estimating the probability density function (PDF) of a random variable based on observed data.

Types:

1. Parametric (e.g., Gaussian model)
2. Nonparametric (e.g., histogram, kernel density)

Importance in classification:

- Helps estimate class probabilities

$P(x \text{ class})$

- Used in Bayesian classification
- Improves decision boundaries
- Useful when distribution is unknown

Q3.) Describe the K-Nearest Neighbor (KNN) Estimation method. How does the choice of K affect performance.

Ans.) KNN estimates the class of a data point based on the majority class among its K nearest neighbors.

Steps:

1. Choose K
2. Compute distance (e.g., Euclidean)
3. Find K nearest neighbors
4. Assign majority class

Effect of K:

- Small K: Low bias, high variance (sensitive to noise)
- Large K: High bias, low variance (smooth decision boundary)

Choosing K :

- Typically odd number
- Selected using cross-validation